

# Allan Tsay

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## SUMMARY

UC Berkeley Computer Science student building practical software across backend, full-stack, and AI-enabled systems. Experienced in translating real operational workflows into usable products, with strong foundations in C/C++, Rust, Python, SQL, operating systems, algorithms, and data structures.

## EDUCATION

**University of California, Berkeley - B.S. Computer Science**

*Expected Winter 2026*

*Coursework: Operating Systems, Algorithms, Data Structures, Computer Architecture, Computer Graphics, Machine Learning, Internet Architecture*

## EXPERIENCE

**Maywei Inc - Intern**

*Summers 2025 & 2026*

- Designed and built an AI-powered internal data assistant for a mobile-phone trading business, giving employees a natural-language interface to inventory, receiving and shipping records, suppliers and customers, and device-level condition data.
- Architected a Python and SQLite backend that uses LLM tool calling to inspect the schema and execute SQL queries, feeding SQL errors back into the workflow for automatic query repair.
- Delivered a React interface that lets nontechnical staff ask questions, inspect structured results, and visualize query output without writing SQL.
- Validated end-to-end query behavior against ground-truth cases and used failures to refine prompting and tool orchestration for common business questions.

## PROJECTS

**Computer Graphics Rasterizer and Geometric Modeling**

*C++*

- Engineered a C++ rendering pipeline for simplified SVG files, combining triangle rasterization and hierarchical transforms to convert vector geometry into raster images while preserving the spatial relationships between nested graphical elements.
- Implemented supersampling-based antialiasing and texture mapping, combining multiple samples per pixel to smooth jagged boundaries and mapping image data onto geometry to render textured surfaces alongside vector shapes.
- Developed Bezier curve and surface evaluation using the de Casteljau algorithm, and implemented Loop subdivision on triangle meshes with a half-edge data structure to represent mesh connectivity and support geometric refinement.

**Pintos OS Kernel**

*C*

- Extended a C-based instructional OS kernel with user-process support, implementing program loading and argument passing to connect kernel-side process initialization with user-space execution and supply programs with their startup inputs.
- Implemented system-call handling to connect user programs with kernel services, enforcing memory protection at the user-kernel interface so applications can invoke OS functionality while preserving the separation of user and kernel memory.

**RISC-V RV32I Emulator**

*Rust*

- Built the memory layer of a RISC-V emulator in Rust, implementing core primitives for representing and accessing emulated memory and connecting memory operations with instruction execution to model program behavior in software.
- Developed an RV32I decode-and-execute pipeline, translating machine-code instructions into software operations and integrating decoding with memory primitives to execute programs according to the instruction set's semantics.

## LEADERSHIP AND ACTIVITIES

**Building Operations Captain - UC Berkeley Student Union**

*Fall 2024 - Present*

- Coordinate daily operations as the first point of contact, supporting 500+ visitors while maintaining policy compliance and facility safety.
- Lead student project teams delivering space improvements and maintain Red Cross First Aid/CPR certification.

**Director - Cal Taiko**

*Aug 2025 - May 2026*

- Directed a 15+ member performing ensemble and managed executive-board planning and coordination.
- Delivered 5+ major campus and community performances, overseeing venue booking, equipment transport, and sound engineering from planning through execution.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, Java, C, C++, Rust, SQL

**Technologies:** React, SQLite, Ollama, Docker, Git, PyTorch

**Spoken Languages:** English and Mandarin Chinese